

LECTURE 1

GROUP AND RING THEORY

1. GROUP ACTIONS

Recall that group theory has its origins from the time of the ancient Greeks, where they studied symmetries of polyhedra. Even though nowadays groups are studied as abstract objects in their own right, we still consider groups as symmetries of (mathematical) objects. For instance, observe that

- S_n permutes $X = \{1, 2, \dots, n\}$;
- the symmetry group G of a cube permutes the cube;
- the multiplicative group $\text{GL}_n(\mathbb{R})$, of invertible $n \times n$ real matrices, acts as linear transformations on $\mathbb{R}^n$. Note that $\text{GL}_n(\mathbb{R})$ is called the *general linear group* over $\mathbb{R}$ of degree n .

We now define a concept that generalises the above examples.

Definition 1.1. Let G be a group and X a set. Then G is said to *act on* X if there is a mapping $\phi : G \times X \rightarrow X$, with $\phi(a, x)$ written $a \cdot x$ (or ax for short), such that for all $a, b \in G$ and $x \in X$,

- (i) $a \cdot (b \cdot x) = (ab) \cdot x$,
- (ii) $e \cdot x = x$.

The mapping ϕ is called an *action* of G on X , and X is said to be a G -set (under the given action).

To be more precise, we have just defined an action *on the left* (or a *left action*). Similarly, a *right action* of G on X is defined to be a mapping $X \times G \rightarrow X$ with $(x, a) \mapsto x \cdot a$ such that $x \cdot (ab) = (x \cdot a) \cdot b$ and $x \cdot e = x$ for all $a, b \in G$ and $x \in X$.

Unless otherwise mentioned, we will only consider groups acting on the left.

Example 1.2. Let $G = D_8 (= \langle x, y \mid x^4 = y^2 = 1, yxy = x^{-1} \rangle)$ and X be the set of vertices $\{1, 2, 3, 4\}$ of a square. The set X is a G -set under the action

$$g \cdot i := g(i), \quad \text{for } g \in D_8 \text{ and } i \in \{1, 2, 3, 4\}.$$

Example 1.3. Let $G = S_5$ and $X = \{x_1, x_2, x_3, x_4, x_5\}$ be a set of beads forming a circular ring. Then X is a G -set under the action

$$g \cdot x_i := x_{g(i)}, \quad \text{for } g \in S_5.$$

Example 1.4. Let G be the additive group $\mathbb{R}$, and $X = \{z \in \mathbb{C} \mid |z| = 1\}$. Then X is a G -set under the action

$$\gamma \cdot c := e^{i\gamma}c, \quad \text{where } \gamma \in \mathbb{R} \text{ and } c \in X.$$

Here the action of γ is the rotation through the angle γ radians, anticlockwise.

Example 1.5. Let G be a group, and let

$$a \cdot x := ax, \quad \text{for } a \in G \text{ and } x \in G.$$

The right-hand side is just the group product. Then, referring to the group axiom of associativity, the set G is a G -set. This action of the group G on itself is called a *(left) translation*.

Example 1.6. Let G be a group, and define

$$a \cdot x := axa^{-1}, \quad \text{for } a \in G \text{ and } x \in G.$$

To see that G is a G -set, let $a, b \in G$. Then

$$\begin{aligned} (ab) \cdot x &= (ab)x(ab)^{-1} = a(bxb^{-1})a^{-1} \\ &= a(b \cdot x)a^{-1} = a \cdot (b \cdot x). \end{aligned}$$

Also $e \cdot x = exe = x$. This proves G is a G -set.

This action of the group G on itself is called *conjugation*.

Example 1.7. Let G be a group acting on a set X . The action of G on X can be extended to the power set $\mathcal{P}(X)$ as follows. We define

$$a \cdot S = \{a \cdot x \mid x \in S\}$$

for all $a \in G$ and $S \subseteq X$. Then, clearly $(ab) \cdot S = a \cdot (b \cdot S)$ and $e \cdot S = S$ for $a, b \in G$ and $S \subseteq X$. Therefore $\mathcal{P}(X)$ is a G -set.

Example 1.8. Let G be a group and let $H < G$. Then the set G/H of left cosets can be made into a G -set by defining

$$a \cdot xH = axH, \quad \text{for } a \in G \text{ and } xH \in G/H.$$

Indeed, for $a, b \in G$ and $xH \in G/H$, we have

$$(ab) \cdot xH = abxH = a \cdot (bxH) = a \cdot (b \cdot xH)$$

and $e \cdot xH = exH = xH$.

Example 1.9. Let G be a group and let $H \trianglelefteq G$. Then the set G/H of left cosets is a G -set if we define

$$a \cdot xH = axa^{-1}H, \quad \text{for } a \in G \text{ and } xH \in G/H.$$

To see this, let $a, b \in G$ and $xH \in G/H$. Then

$$(ab) \cdot xH = abxb^{-1}a^{-1}H = a \cdot bxb^{-1}H = a \cdot (b \cdot xH).$$

Also $e \cdot xH = xH$. Hence G/H is a G -set.

Recall that for a set X , we write S_X (or $\text{Sym}(X)$) for the group of symmetries of X . If $X = \{1, 2, \dots, n\}$, one usually writes S_n (or $\text{Sym}(n)$) instead.

Theorem 1.10. Let G be a group and let X be a set. Then,

- (i) if X is a G -set, then the action of G on X induces a homomorphism $\phi : G \rightarrow S_X$;
- (ii) any homomorphism $\phi : G \rightarrow S_X$ induces an action of G on X .

Proof. (i) We define $\phi : G \rightarrow S_X$ by $(\phi(a))(x) = a \cdot x$, for $a \in G$ and $x \in X$. Clearly $\phi(a) \in S_X$ for $a \in G$. Indeed, it suffices to show that $\phi(a)$ is a bijection, for each $a \in G$. To show that $\phi(a)$ is injective: suppose $a \cdot x = a \cdot y$ for some $x, y \in X$. Then

$$x = a^{-1} \cdot a \cdot x = a^{-1} \cdot a \cdot y = e \cdot y = y.$$

To show that $\phi(a)$ is surjective: consider $x \in X$. Then

$$x = e \cdot x = a \cdot a^{-1} \cdot x = \phi(a)(a^{-1} \cdot x) \in \text{im } (\phi(a)).$$

To show that ϕ is a homomorphism, let $a, b \in G$. Then, recalling that we write ax for $a \cdot x$,

$$\begin{aligned} (\phi(ab))(x) &= (ab)x = a(bx) = a((\phi(b))(x)) = (\phi(a))(\phi(b)(x)) \\ &= (\phi(a)\phi(b))(x) \quad \text{for all } x \in X. \end{aligned}$$

Hence $\phi(ab) = \phi(a)\phi(b)$.

(ii) Define $a \cdot x = (\phi(a))(x)$. Then

$$(ab)x = (\phi(ab))(x) = (\phi(a)\phi(b))(x) = \phi(a)(\phi(b)(x)) = \phi(a)(bx) = a(bx).$$

Also $ex = (\phi(e))(x) = x$. Hence X is a G -set. $\square$

Regarding G as a G -set as in Example 1.5, we obtain the following important result, which was done in Algebraic Structures.

Theorem 1.11 (Cayley's theorem). *Let G be a group. Then G is isomorphic to a subgroup of the symmetric group S_G on the elements of G .*

Proof. By Theorem 1.10(i) there exists a homomorphism $\phi : G \rightarrow S_G$, where $(\phi(a))(x) = ax$, for $a \in G$ and $x \in G$. Suppose $\phi(a)$ equals the identity element in S_G . Then $(\phi(a))(x) = x$ for all $x \in G$. This implies that $ax = x$ for all $x \in X$, and hence $a = e$, the identity in G . Therefore ϕ is injective, and the result follows. $\square$

Further if $H \leq G$ the set G/H of left cosets of H in G is a G -set (cf. Example 1.8). The action of G on G/H gives a more general version of the above theorem, as seen below.

Theorem 1.12. *Let G be a group and $H \leq G$ be a subgroup of index n . Then there is a homomorphism $\phi : G \rightarrow S_n$ such that $\ker \phi = \bigcap_{x \in G} xHx^{-1}$.*

The normal subgroup $\bigcap_{x \in G} xHx^{-1}$ is called the *normal core* of H in G , as is denoted by $\text{core}(H)$.

Proof. Since $|G/H| = n$, we obtain $S_{G/H} \cong S_n$, the symmetric group on n objects. By Theorem 1.10(i) there exists a homomorphism $\phi : G \rightarrow S_n$ such that $\phi(a)(xH) = axH$. Now

$$\begin{aligned} \ker \phi &= \{a \in G \mid \phi(a) = \text{identity permutation}\} \\ &= \{a \in G \mid axH = xH, \forall x \in G\} \\ &= \{a \in G \mid x^{-1}ax \in H, \forall x \in G\} \\ &= \{a \in G \mid a \in xHx^{-1}, \forall x \in G\} \\ &= \bigcap_{x \in G} xHx^{-1}. \end{aligned} \quad \square$$

We obtain the Cayley's Theorem by taking $H = \{e\}$.

Corollary 1.13. *Let G be a group with a normal subgroup H of index n . Then G/H is isomorphic to a subgroup of S_n .*

Proof. This follows from Theorem 1.12, as $H \trianglelefteq G$ implies $\ker \phi = H$. $\square$

We now introduce a subfamily of groups of high importance.

Definition 1.14. A group G is called *simple* if G is non-trivial and the only normal subgroups of G are $\{e\}$ and G .

One can liken simple groups to the “atoms” in the family of groups.

Example 1.15. The groups $\mathbb{Z}_p$, for p a prime, are simple (equivalently, the cyclic groups C_p of order a prime p are simple). The alternating groups A_n , for $n \neq 4$, are simple (see Exercise Sheet 2 for the case $n \leq 4$; the case $n \geq 5$ is more difficult).

Recall that for a group G , a subgroup $H \leq G$ is *proper* if $H \neq G$.

Corollary 1.16. *Let G be a simple group with a proper subgroup of finite index n . Then G is isomorphic to a subgroup of S_n .*

Proof. This likewise follows from Theorem 1.12 by recalling that $\ker \phi \trianglelefteq G$. □

Definition 1.17. Let G be a group acting on a set X , and let $x \in X$. Then the set

$$G_x := \{g \in G \mid gx = x\},$$

is called the *stabiliser* of x in G .

The stabiliser G_x is actually a subgroup; see Exercise Sheet 1.

Example 1.18. If G acts on itself by conjugation (cf. Example 1.6), then for $x \in G$, the stabiliser of x in G

$$G_x = \{a \in G \mid axa^{-1} = x\} = C_G(x),$$

coincides with the centraliser $C_G(x)$ of x in G .

Example 1.19. Let $G = D_8$ act on the square with vertices 1, 2, 3, 4. If $x = 1$, then $G_x = \{e, \text{ the reflection along the diagonal through the vertex 1}\}$.

Example 1.20. Consider the G -set G/H , where $H < G$ (cf. Example 1.8). Here the stabiliser of a left coset xH is the subgroup

$$\begin{aligned} \{g \in G \mid gxH = xH\} &= \{g \in G \mid x^{-1}gx \in H\} \\ &= \{g \in G \mid g \in xHx^{-1}\} \\ &= xHx^{-1}. \end{aligned}$$

Definition 1.21. Let G be a group acting on a set X , and let $x \in X$. Then the set

$$Gx := \{ax \mid a \in G\}$$

is called the *orbit* of x in G .

One can also say the G -orbit of x and also write $G \cdot x$.

Example 1.22. Consider G acting on itself as in Example 1.5. The orbit of $x \in G$ is $Gx = \{ax \mid a \in G\} = G$.

If there is only one G -orbit, we say that the action is *transitive*.

Example 1.23. Consider G as a G -set as in Example 1.6, where the action of the group G is by conjugation. The orbit of $x \in G$ is $Gx = \{axa^{-1} \mid a \in G\}$, which is called the *conjugacy class* of x and denoted by $\text{ccl}(x)$.

If G is abelian, then the conjugacy class of $x \in G$ is the singleton $\{x\}$. See Example 1.26 below for a non-trivial example.

We now prove an important result.

Theorem 1.24 (Orbit-Stabiliser Theorem). *Let G be a group acting on a set X . Then the set of all orbits in X under G is a partition of X . Furthermore, for each $x \in X$ there is a bijection $Gx \rightarrow G/G_x$ and hence*

$$|Gx| = |G : G_x|.$$

Therefore, if X is a finite set,

$$|X| = \sum_{x \in C} |G : G_x|,$$

where C is a subset of X containing exactly one element from each orbit.

Proof. For any $x, y \in X$, let $x \sim y$ mean that $x = ay$ for some $a \in G$. Then $\sim$ is an equivalence relation on X . Indeed, we have $x = ex$ for all $x \in X$, so $x \sim x$. If $x = ay$, then $y = a^{-1}x$, so we have $x \sim y$ implies $y \sim x$. If $x = ay$ and $y = bz$, then $x = (ab)z$, that is, if $x \sim y$ and $y \sim z$, we have $x \sim z$. Hence the equivalence class of $x \in X$ is the orbit Gx . Thus the set of all orbits is a partition of X , as equivalence classes form a partition. Therefore,

$$X = \bigcup_{x \in C} Gx \quad (\text{disjoint}),$$

where C is any subset of X containing exactly one element from each orbit.

Given $x \in X$, consider the mapping

$$\phi : Gx \rightarrow G/G_x \quad \text{given by} \quad ax \mapsto aG_x \quad \text{for all } a \in G.$$

For any $a, b \in G$,

$$ax = bx \iff a^{-1}bx = x \iff a^{-1}b \in G_x \iff aG_x = bG_x.$$

Hence ϕ is well defined and injective. Moreover ϕ is obviously surjective and therefore bijective. Hence $|Gx| = |G/G_x| = |G : G_x|$.

Since X is the disjoint union of orbits Gx , it follows that if X is finite, then

$$|X| = \sum_{x \in C} |Gx| = \sum_{x \in C} |G : G_x|. \quad \square$$

The partition of X given by the orbits in X under G is called the *orbit decomposition of X under G* . Regarding the final statement, the corresponding statement holds if G is infinite, only one needs to evoke the Axiom of Choice.

In the next theorem, we give a version of Theorem 1.24 for the case when G acts on itself by conjugation.

Theorem 1.25. *Let G be a group. Then,*

- (i) *the set of conjugacy classes of G is a partition of G ;*
- (ii) *$|\text{ccl}(a)| = |G : C_G(a)|$ for $a \in G$;*
- (iii) *if G is finite, then $|G| = \sum |G : C_G(a)|$ with the element a running over exactly one element from each conjugacy class.*

As this result just follows from the Orbit-Stabiliser Theorem, there is no need to give a proof, but if one wanted to prove it from scratch, here is a proof.

Proof. (i) Define a relation $\sim$ on G as follows:

$$a \sim b \quad \text{if} \quad a = bx^{-1} \quad \text{for some } x \in G.$$

Clearly $\sim$ is an equivalence relation on G . The equivalence class of a in G is then the set $\{xax^{-1} \mid x \in G\}$, which is also the conjugacy class $\text{ccl}(a)$ of a . This proves $G = \bigcup \text{ccl}(a)$, a disjoint union of conjugacy classes.

(ii) Let $a \in G$. Then the mapping

$$\sigma : \text{ccl}(a) \rightarrow G/C_G(a) \quad \text{given by} \quad xax^{-1} \mapsto xC_G(a)$$

is trivially onto. Also it is one-to-one. To see this, let $x, y \in G$. Then

$$xC_G(a) = yC_G(a) \implies y^{-1}x \in C_G(a) \implies y^{-1}xa = ay^{-1}x \implies xax^{-1} = yay^{-1}.$$

Hence σ is a bijection, proving $|\text{ccl}(a)| = |G/C_G(a)|$.

(iii) Follows from (i) and (ii). $\square$

The partition of G given by the conjugacy classes in Theorem 1.25(i) is called the *class decomposition* of G . The equation in part (iii) is called the *class equation* (or *class equation formula*) of G .

Example 1.26. Let $G = S_3$. A direct check yields that the conjugacy classes are:

$$\begin{array}{ll} \{e\} & \\ \{(1\ 2), (1\ 3), (2\ 3)\} & \text{transpositions} \\ \{(1\ 2\ 3), (1\ 3\ 2)\} & \text{3-cycles} \end{array}$$

If we take $C = \{e, (1\ 2), (1\ 2\ 3)\}$, then

$$\begin{array}{ll} C_G(e) = G = S_3 & |S_3 : S_3| = 1, \\ C_G((1\ 2)) = \{e, (1\ 2)\} = \langle (1\ 2) \rangle & |S_3 : \langle (1\ 2) \rangle| = 3, \\ C_G((1\ 2\ 3)) = \{e, (1\ 2\ 3), (1\ 3\ 2)\} = \langle (1\ 2\ 3) \rangle = A_3 & |S_3 : A_3| = 2. \end{array}$$

The class equation is then

$$|S_3| = 3! = 6 = 1 + 3 + 2.$$

Let G be a group, and let S be a subset of G . If $x \in G$, then the set

$$x^{-1}Sx = \{x^{-1}sx \mid s \in S\}$$

is called a *conjugate* of S . This was already seen in Example ???. One also uses the notation S^x for $x^{-1}Sx$.

Definition 1.27. Let S and T be two subsets of a group G . Then T is said to be *conjugate* to S if there exists $x \in G$ such that $T = x^{-1}Sx$.

Clearly “being conjugate to” is an equivalence relation in the power set $\mathcal{P}(G)$ of G .

Let G be a group. For any subset S of G , the *normaliser* of S in G is the set

$$N_G(S) = \{x \in G \mid x^{-1}Sx = S\}.$$

Note that if $S = \{x\}$ consists of only one element, one says the *centraliser* of x in G .

Following the method of the proof of Theorem 1.25 or by specialising Theorem 1.24 to the G -set $\mathcal{P}(G)$, where G acts on $\mathcal{P}(G)$ by conjugation, we obtain:

Theorem 1.28. *Let G be a group. Then for any subset S of G ,*

$$|\text{ccl}(S)| = |G : N_G(S)|.$$

Example 1.29. Let H be a subgroup of a finite group G . Let $A, B \in \mathcal{P}(G)$, the power set of G . Define A to be conjugate to B with respect to H as $B = hAh^{-1}$ for some $h \in H$. Then show that

(i) the conjugacy defined in $\mathcal{P}(G)$ is an equivalence relation;

- (ii) if $\text{ccl}_H(A)$ is the equivalence class of $A \in \mathcal{P}(G)$ (that is, the conjugacy class of A with respect to H), then

$$|\text{ccl}_H(A)| = |H : H \cap N_G(A)|.$$

Solution: This follows from the Orbit-Stabiliser Theorem (Theorem 1.24) by taking X to be the power set $\mathcal{P}(G)$ of G and H to be the group that acts on X by conjugation. Then the orbit of $A \in X$ is the conjugacy class $\text{ccl}_H(A)$, and the stabiliser is $H \cap N_G(A)$.

[Alternatively, one can use a direct argument as follows: for (i), it is a straightforward check. For (ii), consider the mapping

$$\sigma : \text{ccl}_H(A) \rightarrow H/(H \cap N_G(A)),$$

where

$$\sigma(hAh^{-1}) = h(H \cap N_G(A)), \quad \text{for } h \in H.$$

The map σ is clearly onto. Also σ is one-to-one. Indeed, let $h_1, h_2 \in H$. Then

$$\begin{aligned} h_1(H \cap N_G(A)) = h_2(H \cap N_G(A)) &\implies h_1^{-1}h_2 \in H \cap N_G(A) \\ &\implies h_1^{-1}h_2A = Ah_1^{-1}h_2 \\ &\implies h_1Ah_1^{-1} = h_2Ah_2^{-1}. \end{aligned}$$

Hence σ is bijective.]